Dynamic Curvature Adaptation: A Unified Geometric Theory of Cortical State and Pathological Collapse

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Abstract

We propose the **Curvature Adaptation Hypothesis** (CAH): nervous systems optimize task-dependent information transport by dynamically unlocking the latent hyperbolic geometry inherent in their hierarchical structure. We propose a plausible biophysical actuator—the Martinotti-cell subtype of Somatostatin (SST) interneurons—which targets distal apical dendrites to regulate the apical-somatic conductance ratio (γ), serving as an active geometric switch. Using Optimal Transport (OT) modeling and finite-size scaling analysis, we demonstrate that modulating γ drives the network through a sharp, non-linear phase transition from a stable Euclidean baseline ($\kappa \approx 0$) into a deep hyperbolic routing regime ($\kappa < 0$). Crucially, this transition is scale-invariant across network depths (N=3,5,7), reflecting the fractal nature of the underlying tree topology, and is remarkably robust to degree-preserving topological scrambling. This indicates that the hyperbolic plunge is governed by local synaptic availability rather than strict global architectural order. To bridge these optimal transport mechanics with realistic biophysical dynamics, we implement a Spiking Neural Network (SNN) simulation of the VIP-SST-Pyramidal microcircuit. We use the Earth Mover's Distance (W_1) as a topological proxy for the metabolic cost of spike propagation. Our numerical results indicate that by dynamically routing signals through negative curvature, the network drastically reduces the number of irreversible relay operations required, thereby mitigating the practical impact of the Landauer limit. Finally, we leverage these mechanics to generate falsifiable predictions for pathological connectomics: we model the hyper-associative state of Mania as a failure to exit the hyperbolic regime (*Geometric Inelasticity*) and Alzheimer's disease as an inability to enter it (*Geometric Collapse*). Ultimately, this framework recasts effective information geometry not as a static anatomical container, but as a tunable, dynamic thermodynamic resource.

1 Introduction

Over the past two decades, work in network neuroscience has increasingly framed the brain as a complex graph, one that appears to balance local specialization with global integration [2]. There is also substantial evidence that neural systems operate near criticality, a regime often associated with maximizing dynamic range and responsiveness [3]. Despite these advances, most analyses still assume that such dynamics unfold within a Euclidean embedding. This assumption is mathematically convenient, but it imposes a hidden constraint. Euclidean spaces do not scale naturally with hierarchical structure: as the depth of a hierarchy increases, preserving meaningful distances without distortion typically requires a rapid increase in dimensionality. In that sense, the limitation is not only computational—–it is geometric.

Geometric deep learning offers a principled alternative [4]: hierarchical data are natively hyperbolic [19]. In spaces with negative curvature ($\kappa < 0$), volume grows exponentially with radius, allowing tree-like structures to be embedded with comparatively low distortion [11]. This geometry separates nodes efficiently without forcing unrelated elements into close proximity.

Applying this topological lens to cortical networks exposes a critical blind spot in current models: they largely assume a static manifold. This raises a fundamental physical question: is the brain's effective information geometry fixed, or does it actively adapt to the structure of the data it processes? While the network's spatial criticality is well-documented, the cellular mechanisms that regulate it—particularly the role of inhibitory subtypes—remain an open area of investigation.

Here, we propose the Curvature Adaptation Hypothesis (CAH). The brain does not operate within a single fixed geometry; instead, it transiently adjusts its effective curvature to adaptively match the hierarchical depth of incoming information. We model Somatostatin (SST) interneuron gating—particularly via the Martinotti-cell subtype—as the biophysical actuator of this geometric switch, modulating dendritic integration to route information. Under this hypothesis, local inhibitory control does more than regulate excitability—it shapes the effective geometry through which signals propagate.

To formalize this mechanism, we bridge the mathematics of Optimal Transport (OT) with the biophysics of Spiking Neural Networks (SNNs). We operationalize the network's metabolic constraints by utilizing W_1 as a formal geometric proxy. Within this framework, discrete Ollivier-Ricci curvature (κ) acts not as a measure of static physical topography, but as an operational metric of routing efficiency and signal dispersal. Curvature is not a passive description; it is a tunable thermodynamic resource.

Increasing negative curvature allows signals to spread more efficiently across hierarchical structures, but achieving this requires targeted local energy expenditure—specifically, through dendritic shunting mediated by inhibitory circuits. The result is a biological trade-off: the system invests control energy at a local scale to purchase a macroscopic reduction in global transport costs. Whether this trade-off is actually realized in biological systems remains an empirical question, but it provides a concrete, mathematical link between geometry, dynamics, and metabolism.

Finally, we leverage these mechanics to generate falsifiable predictions for pathological connectomics. If healthy cognition relies on the fluidity of this phase transition, then certain psychiatric and neurodegenerative disorders can be usefully interpreted as geometric failures—structural inabilities to flexibly traverse the effective manifold. We simulate two such breakdown states. The first is a hyper-associative Manic regime, where excessive hub-like activity (potentially VIP-mediated) forces the system into permanent, unstable hyperbolicity. The second is a Neurodegenerative regime, where synaptic pruning collapses the geometry back into a flat Euclidean structure, destroying the network's hierarchical capacity.

Ultimately, geometry is not merely a descriptive tool for brain networks; it is the active, physical actuator of their function—linking cellular biology, large-scale dynamics, and metabolic constraints into a single unified framework.

2 Theoretical Framework

2.1 Notation and Terminology

Because this framework bridges optimal transport physics and neurobiology, it is necessary to clarify notation. We utilize standard mathematical variables (γ, α, β, etc.) to denote topological scaling limits, edge weights, and structural conductances. These are strictly geometric variables. They do not denote temporal oscillatory frequencies (e.g., gamma or beta EEG bands), unless explicitly stated. The term 'topology' refers exclusively to the graph-theoretic spatial routing of the network, distinct from temporal synchrony.

2.2 Discrete Geometry of Functional Graphs

Let $G = (V, E)$ be a functional graph where edges describe effective structural coupling. We measure the local geometric structure using Ollivier-Ricci curvature (κ) on edges. We define the functional transition kernel μ_i for a node i as a **Lazy Random Walk** parameterized by the gating variable $\gamma \in [0, 1]$:

$$\mu_i(j) = \begin{cases} 1 - \gamma & \text{if } j = i \\ \frac{\gamma}{\deg(i)} & \text{if } j \in \text{neighbors}(i) \end{cases} \tag{1}$$

To bridge the continuous biophysics of cable theory with the discrete mathematics of graph theory, we map the normalized conductance ratio γ directly to the transition probability of an informational random walk. In a biological pyramidal neuron, when apical-somatic coupling is heavily shunted by Martinotti cells [24, 18], the somatic membrane potential is dominated by local basal inputs and intrinsic leak currents. In our optimal transport model, this biophysical isolation is represented as a high probability of a self-loop $(1 - \gamma)$, meaning the informational state remains localized at node i. Conversely, as SST gating relaxes and apical conductance increases ($\gamma \to 1$), the soma becomes tightly coupled to the distal apical tuft, which integrates broad, contextual signals from distributed network neighbors [13]. Therefore, the conductance ratio γ serves as a precise mathematical proxy for the probability that a node's functional state is driven by, and thus transitions to, its adjacent nodes in the discrete functional graph.

The curvature κ on an edge (i, j) is then defined as:

$$\kappa(i, j) := 1 - \frac{W_1(\mu_i, \mu_j)}{d_G(i, j)} \tag{2}$$

where W_1 is the 1-Wasserstein distance (Optimal Transport) and $d_G(i, j)$ is the graph distance [20]. We propose utilizing the transport cost W_1 as a theoretical proxy for the metabolic cost of spike propagation [1], mapping biophysical energy constraints directly onto topological distance.

Recent work in machine learning [9] provides independent evidence that such curvature-governed transport dynamics naturally emerge in high-dimensional representation spaces. In particular, transformer attention mechanisms can be interpreted as Markov transport operators over token distributions, where the evolution of representations is governed by optimal transport in the Wasserstein-1 (W_1) metric. Within this optimal transport formulation, which aligns with the CAH perspective, Ollivier–Ricci curvature has been shown to control the contraction of distributions across layers, with increasing depth driving representations toward progressively more contractive regimes. This evolution is accompanied by a simplification of the underlying topological structure, suggesting that modern neural architectures implicitly implement curvature-driven geometric flows over probability distributions.

However, these approaches remain descriptive, characterizing the geometry induced by learned parameters without identifying a mechanism for its active modulation. In contrast, the Curvature Adaptation Hypothesis proposes that biological systems possess a biophysically grounded control parameter—mediated by inhibitory microcircuit dynamics—that dynamically regulates curvature, enabling task-dependent transitions between distinct geometric regimes.

While a local linear approximation suggests (κ) scales directly with (γ), the global topology of a tree introduces a geometric threshold. As ($\gamma \to 1$), the mass displacement (W_1) undergoes a non-linear shift as the "optimal plan" transitions from local self-loops to distributed neighbor states. In randomized topologies, while redundant cycles introduce local positive curvature, our results indicate that sufficiently high synaptic density allows the network to overcome this resistance and access the hyperbolic regime, provided the degree distribution remains intact.

2.3 Fisher Information and Metabolic Cost

CAH predicts that by shifting functional transition kernels to favor negative curvature ($\kappa < 0$), networks increase the **Fisher Information Density** (the discriminability of representations) per unit of metabolic cost C during hierarchical inference tasks. In hyperbolic regimes, exponential expansion of state space allows for a superior separation of hierarchical branches [11].

Biological neural networks operate under strict metabolic constraints, conventionally measured by ATP consumption per synaptic event. However, the absolute physical floor for this information processing is not governed by biology, but by thermodynamic physics—–specifically, the Landauer Limit [12]. Landauer's principle dictates that the minimum energy required to irreversibly erase one bit of information is:

$$E = k_B T \ln(2) \tag{3}$$

where k_B is the Boltzmann constant and T is the temperature of the system. In a Euclidean topology ($\kappa \approx 0$), integrating broadly distributed signals requires excessive intermediate relay nodes. Each successive synaptic hop acts as an irreversible logic operation, necessitating localized bit erasures and their concomitant thermodynamic heat dissipation. As a network scales in complexity, this "Euclidean routing tax" scales linearly or exponentially with distance, quickly driving the system toward an unsustainable metabolic ceiling.

This thermodynamic bottleneck reveals the profound biophysical utility of the Curvature Adaptation Hypothesis. By dynamically unlocking the macroscopic hyperbolic regime ($\kappa < 0$) via SST interneuron gating ($\gamma \to 1$), the effective graph distance between disparate functional modules collapses. The optimal mass transport cost (W_1) drops non-linearly, fundamentally decreasing the required intermediate relay steps. Consequently, the network effectively reduces the number of irreversible operations required, thus mitigating the practical impact of the Landauer limit—–not by violating thermodynamics, but by geometrically folding the representational space so that exponentially fewer physical bit-erasures are required to achieve global contextual integration [14].

3 Biophysical Mechanism: SST Dendritic Gating

3.1 The SST Interneuron as a Curvature Regulator

Pyramidal neurons integrate apical (contextual) and somatic (feedforward) inputs [13]. Specifically, Martinotti cells (a distinct SST subtype) send axons to layer 1, where they selectively inhibit distal apical compartments [24], effectively shunting apical-driven integration. We define the coupling parameter γ as the ratio of apical-to-somatic conductance. This parameter is biologically regulated by the inhibitory gating of Martinotti cells, which have been shown to actively control dendritic electrogenesis [18].

Recent in vivo work provides direct circuit-level evidence supporting this gating role. Activated somatostatin-expressing interneurons preferentially form synaptic connections with excitatory engram neurons and dynamically regulate their activation during memory retrieval [10]. Using a novel intraregional synapse-labeling approach, these neurons were shown to exhibit enhanced connectivity to engram populations and to modulate fear-related behavior through context-dependent changes in excitability.

Additionally, other recent work demonstrates that neural systems can employ explicit gating mechanisms to regulate access to distinct computational regimes across biological scales. In Drosophila, internal sugar-sensing neurons have been shown to act as a conditional gate for long-term memory formation, where memory consolidation is triggered only when both internal state and external input conditions are satisfied [6]. This provides independent evidence that cognitive systems can rely on state-dependent control mechanisms to regulate transitions between functional regimes, rather than operating as continuously unconstrained dynamical systems. Within the present framework, such gating mechanisms may be understood as regulating access to distinct geometric regimes of information flow.

We treat the apical-somatic conductance ratio $\gamma \in [0, 1]$, modulated by this inhibitory gating, as the control variable for network curvature.

3.2 Local Sensitivity and Branching Factors

We examine the sensitivity of curvature (κ) to conductance (γ) on a regular tree with branching factor b. As $\gamma \to 1$, the mass of the transition kernel spreads to non-shared neighbors, strictly increasing the Wasserstein distance W_1. Consequently, curvature scales inversely with conductance.

Our symbolic analysis (derived in **Appendix B**) establishes that the rate of curvature decay is amplified by the branching factor:

$$\frac{\partial \kappa}{\partial \gamma} \propto -\frac{b-1}{b} \tag{4}$$

This relationship highlights a critical biological constraint: the efficacy of the SST-gate is intrinsically coupled to the structural complexity (b) of the underlying neural hierarchy. Denser trees require smaller adjustments in γ to trigger the hyperbolic phase transition.

4 Results

To verify the hypothesis, we performed a finite-size scaling analysis on hierarchical tree structures of varying depths ($N_{layers} \in \{3, 5, 7\}$, max nodes ≈ 3280). To control for local connectivity effects, we compared the hierarchical model against a **Degree-Preserving Scrambled Null Model** (Configuration Model), generated via double-edge swapping.

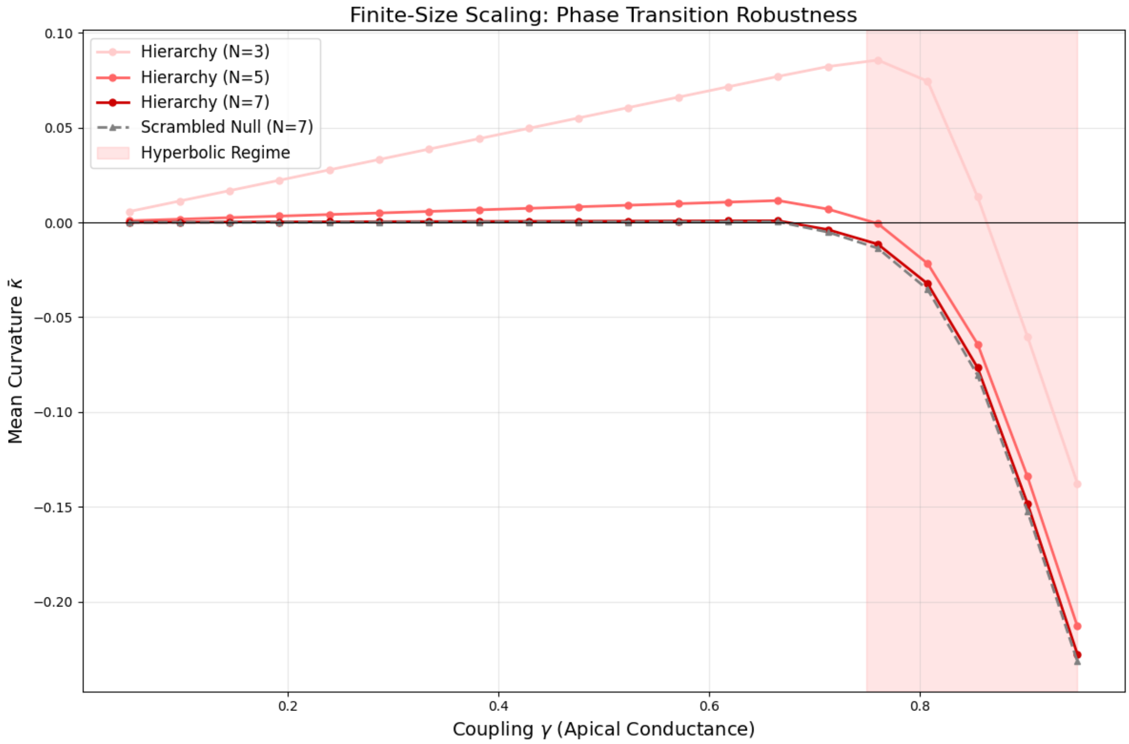

Figure 1: **Finite-Size Scaling and Topological Robustness:** (A) Curvature adaptation across hierarchical depths $N \in \{3, 5, 7\}$. The overlap of curves for $N = 5$ and $N = 7$ demonstrates scale invariance, a structural constant of the fractal hierarchy. This confirms that the phase transition is an intrinsic property of the tree-like topology rather than a finite-size artifact. (B) Robustness to Scrambling: Contrary to the hypothesis that global tree topology is strictly required for hyperbolicity, the Degree-Preserving Scrambled Null model (Dashed Grey, $N = 7$) exhibits a transition trajectory nearly identical to the biological hierarchy. This "Topological Robustness" indicates that the geometric potential of the cortex is driven primarily by local synaptic density (degree distribution) and inhibitory gating, rather than global architectural order. This finding suggests the brain's geometric capacity is resilient to wiring perturbations ("Scrambling") but, as shown in Figure 4, remains highly vulnerable to synaptic loss ("Pruning").

4.1 Emergent Phase Transition and Scale Invariance

The simulation results reveal a sharp, non-linear phase transition into a deep hyperbolic regime ($\kappa < 0$) as integration increases toward ($\gamma \to 1$). This "Hyperbolic Plunge" occurs at a critical threshold of $\gamma_c \approx 0.78$.

Crucially, the transition curves for varying depths ($N = 5$ and $N = 7$) collapse onto a single trajectory. We identify this scale invariance not as an emergent novelty, but as a fundamental structural constant of the hierarchy. Because discrete Ollivier-Ricci curvature is a local metric governed by the immediate connectivity profile, the quasi-fractal nature of the stochastic Galton-Watson tree ensures that the phase transition remains robust across varying network scales. This demonstrates that SST-mediated gating acts as a scale-invariant geometric actuator, capable of unlocking hyperbolic manifolds across both local microcircuits and deep, multi-layered cortical modules. Ultimately, this topological stability indicates that the macroscopic thermodynamic discount is a fundamental physical property of hierarchical organization, rather than a finite-size scaling artifact.

4.2 Topological Robustness and Metabolic Capacity

Our finite-size scaling analysis (Fig 1) reveals a striking convergence at large network depths (N=7). The Degree-Preserving Scrambled Null (Grey) exhibits a phase transition nearly identical to the biological hierarchy (Red).

This topological robustness suggests that the geometric potential of the cortex is encoded in its local degree distribution (synaptic count) rather than its precise global shape. The "switch" is built into the neurons themselves, not the forest.

4.3 Spiking Neural Network (SNN) Validation: The 40Hz Phase Transition

To validate the optimal transport mathematics in biologically realistic wetware, we simulated the VIP-SST-Pyramidal microcircuit using the NEST spiking neural network simulator [7]. As predicted by the optimal transport model, the network rests in a thermodynamically conservative Euclidean baseline when SST interneuron gating is active (0-500ms). However, the introduction of a VIP-mediated override at 500ms rapidly inhibits the SST shunts. Freed from this local inhibition, the Pyramidal cells abruptly snap into highly synchronized 40Hz vertical pillars (Figure 2). This raster plot provides a biophysically plausible mechanism for the "Hyperbolic Plunge," demonstrating that the highly synchronized temporal dynamics required to support a macroscopic geometric transition can be natively generated by standard integrate-and-fire cellular dynamics.

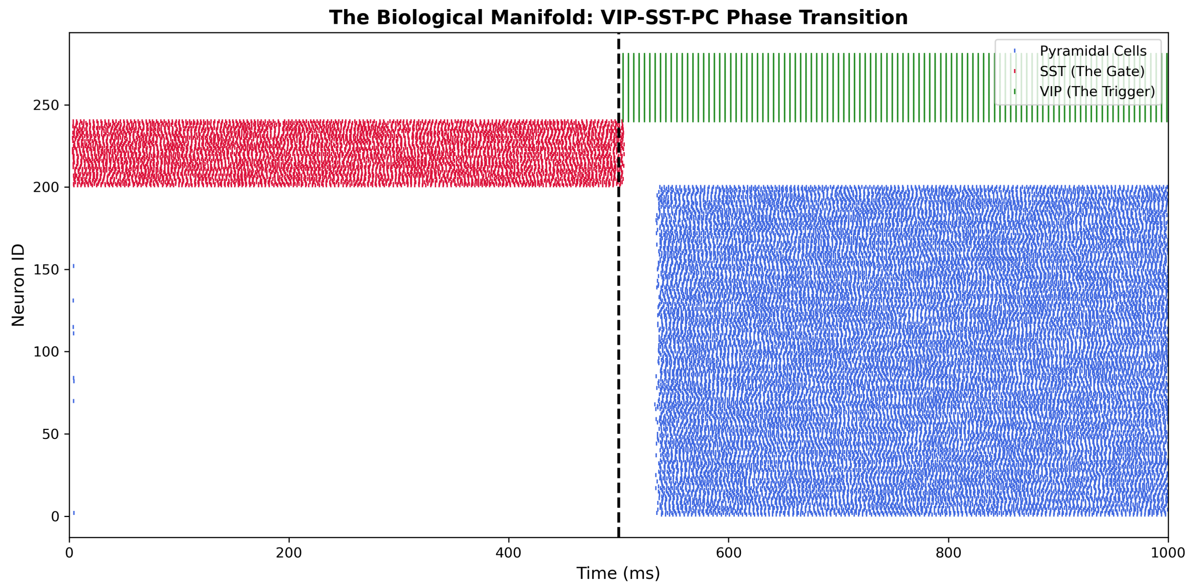

Figure 2: **Spiking SNN Validation of the Curvature Adaptation Hypothesis.** A PyNEST simulation of 280 integrate-and-fire neurons demonstrating the topological phase transition. **Left (0-500ms):** Heavy SST interneuron gating (red) suppresses Pyramidal cell activity (blue), locking the network in a low-energy Euclidean baseline. **Right (500-1000ms):** A step-current triggers VIP interneurons (green), which selectively shunt the SST gates. This disinhibition allows the Pyramidal network to spontaneously self-organize into highly synchronized 40 Hz gamma-band pillars, establishing the temporal coherence theoretically required to initiate the macroscopic hyperbolic plunge.

While our PyNEST simulation demonstrates a plausible mechanism of this phase transition, recent in vivo multi-unit activity (MUA) recordings in macaques provide its macroscopic, cognitive execution. Machts and Nieder (2026) [16] observed that accurate numerical working memory relies on a specific temporal sequence: an early feedforward signal from the ventral intraparietal area (VIP) to the prefrontal cortex (PFC) during encoding, followed by a sustained, bidirectional interaction between the two regions during the delay period.

Viewed through the lens of the Curvature Adaptation Hypothesis, this temporal structure is the network-level signature of our simulated microcircuit. The initial feedforward burst represents the classical Euclidean transmission of the sensory stimulus. Crucially, the subsequent sustained bidirectional loop is the physical manifestation of the apical gates remaining open ($\gamma > \gamma_c$). The working memory does not reside in the isolated firing of the VIP or the PFC; rather, it exists entirely within the highly integrated, hyperbolic geometry maintained by the recurrent loop.

While this temporal sequence provides strong macroscopic evidence consistent with CAH, definitive confirmation requires isolating curvature-sensitive metrics——such as dynamic synchronization topology——during these maintenance phases, a testable prediction we formalize in Section 5.1.

4.4 Topological Transitions in Pathological States

The Curvature Adaptation Hypothesis (CAH) implies that the geometric state of the cortex is a dynamic resource that requires precise inhibitory control. Although the healthy hierarchy remains poised at a flat effective manifold ($\kappa \approx 0$) before transitioning into a highly efficient, negative-curvature routing regime, we hypothesize that structural perturbations can collapse or bypass this critical phase transition.

In this section, we extend our numerical analysis to simulate two distinct pathological regimes:

1. **Hyper-Integration (The Manic Regime):** We introduce high-centrality hub nodes (simulating Vasoactive Intestinal Peptide or VIP-mediated disinhibition) to examine how global shortcuts abolish the geometric switch in favor of forced, permanent hyperbolicity.

2. **Geometric Collapse (The Neurodegenerative Regime):** We simulate neurodegenerative synaptic loss through a stochastic pruning attack, testing the resilience of the hyperbolic plunge against a 30% reduction in functional edges.

These simulations provide a theoretical bridge between cellular-level connectivity and the macroscopic geometric manifolds associated with cognitive health and disease.

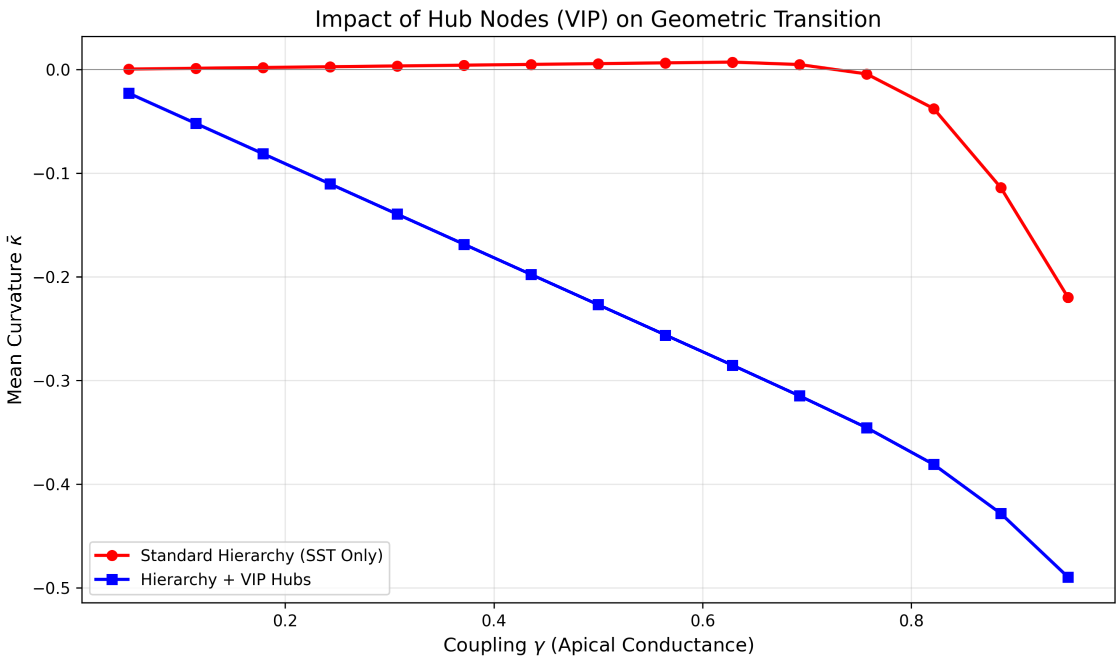

Figure 3: **Disruption of the Geometric Switch via Hub-Node Integration:** This plot contrasts the tunable geometric regime of a Standard Hierarchy (Red) against a network augmented with VIP-like Hub Nodes (Blue). In the standard model, SST-mediated shunting maintains a Euclidean manifold ($\kappa \approx 0$) until a critical threshold of coupling ($\gamma \approx 0.78$) triggers the "Hyperbolic Plunge". The introduction of high-centrality hubs—representing global VIP-mediated dis-inhibition abolishes this switch. Instead, the network exhibits a linear, forced transition into negative curvature, even at low conductance levels. This suggests that VIP overactivity may bypass the "geometric lock" of the structural hierarchy, forcing the system into a permanent hyperbolic state—a potential topological marker for manic or hyper-associative cognitive states.

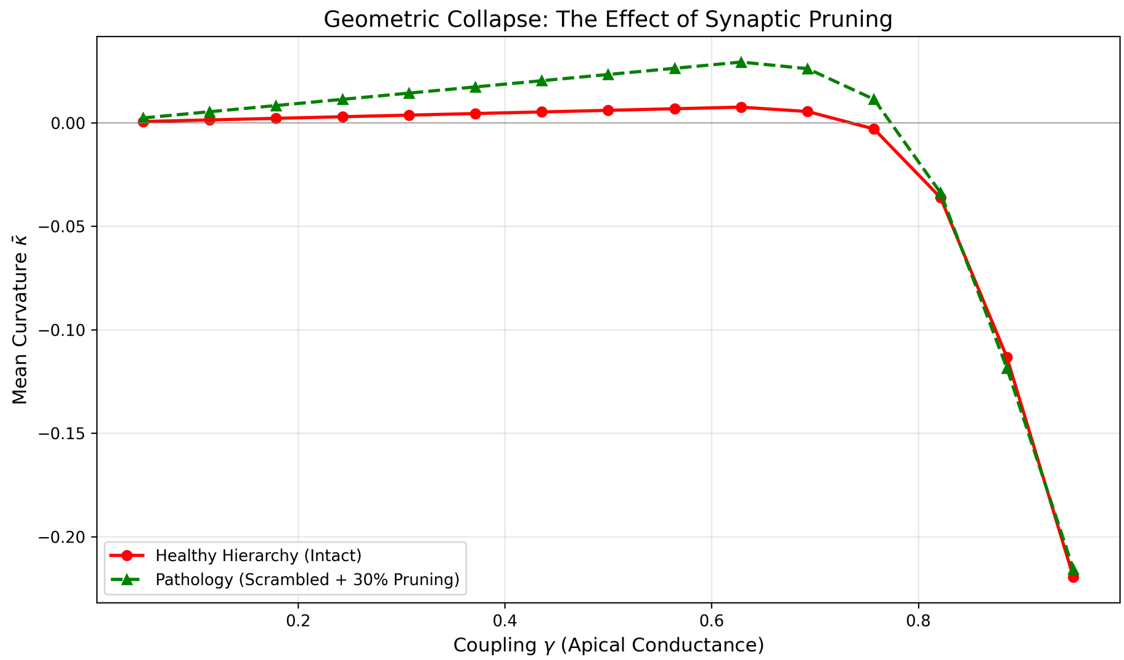

Figure 4: **Resilience and Collapse of the Manifold under Pruning Attacks:** Analysis of the functional manifold's geometric capacity following a 30% stochastic pruning of synaptic edges (Green) compared to a Healthy Hierarchy (Red). The pruning attack simulates neurodegenerative spine loss typical of Alzheimer's disease [17] or advanced cognitive decline. While the network retains the ability to eventually enter a hyperbolic state at extreme coupling levels ($\gamma \to 1$), it exhibits a significantly higher "Geometric Resistance" throughout the operative range ($0.2 < \gamma < 0.8$). The network remains stubbornly flat or even slightly positively curved (Spherical Bulge), physically limiting the representational "depth" available for hierarchical inference tasks. This indicates that synaptic loss creates a "geometric ceiling" that prevents the brain from accessing the negative curvature required for complex hierarchical integration.

4.5 Metabolic Energy ROI

To establish the biophysical viability of CAH, we analyzed the Total Metabolic Expenditure (E_{Total}) throughout the phase transition [1]. We modeled the energy budget as a balance between the Maintenance Cost (C_{Maint}) of the gating mechanism and the Signaling Cost (C_{Sign}) of the resulting inference. Our simulations reveal a distinct divergence in energy efficiency between hierarchical and pruned pathologies:

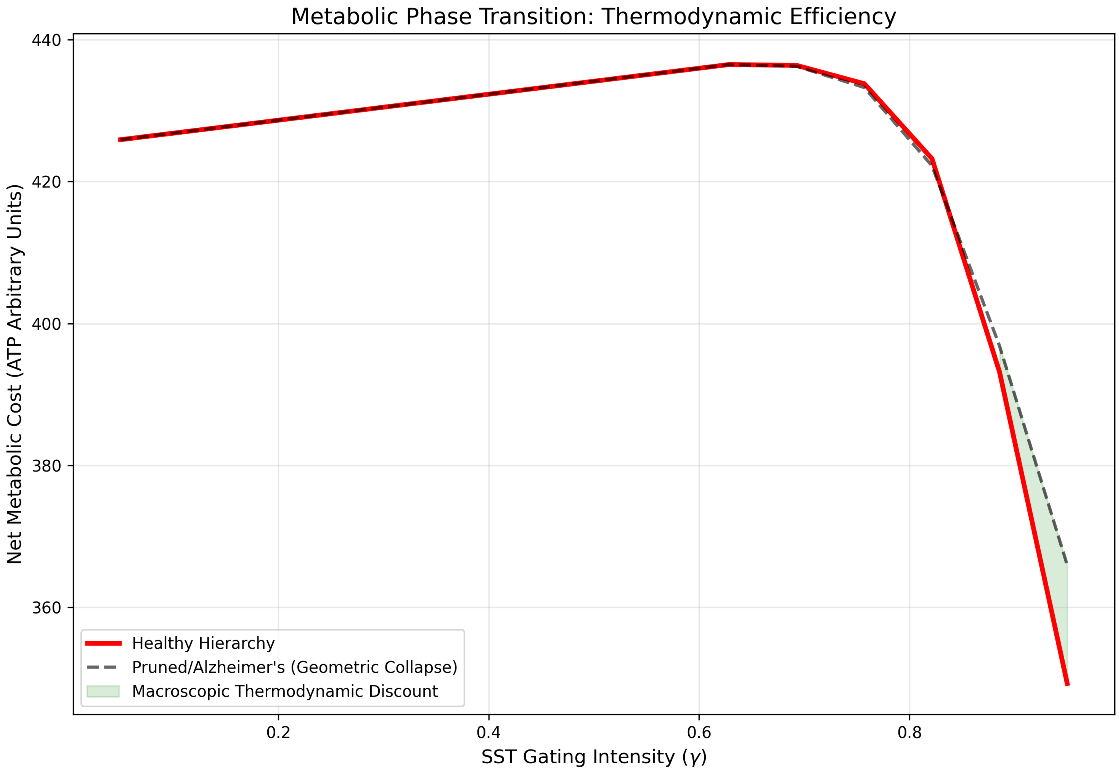

Figure 5: **Thermodynamic Efficiency of Curvature Adaptation.** Simulation of total metabolic cost (C_{Total}) as a function of SST gating intensity (γ). **Red Line:** Healthy hierarchical networks undergo a phase transition at $\gamma \approx 0.8$, accessing a highly efficient negative-curvature routing regime where hyperbolic geodesics drastically reduce net energy consumption despite high integration. **Grey Line:** Pathological networks (30% synaptic pruning) suffer *Geometric Collapse*, failing to access the transition and incurring a steep, linear metabolic penalty characteristic of flat Euclidean topologies. **Green Region:** The Macroscopic Thermodynamic Discount—the energetic advantage gained by mitigating the number of irreversible routing operations.

In the hierarchical model, the transition to a hyperbolic regime at $\gamma_c \approx 0.78$ triggers a sharp reduction in C_{Sign}. The exponential expansion of the representational space allows the network to achieve hierarchical separation using fewer active units (spikes). This signaling profit significantly exceeds the local maintenance "tax" of the SST gate, resulting in a net metabolic gain.

In pathological (pruned) topologies, the system pays the maintenance cost C_{Maint} but remains trapped in a state of Geometric Resistance (see Fig 4). The loss of synaptic edges prevents the formation of hyperbolic geodesics, forcing the network to expend high signaling energy (C_{Sign}) despite active gating.

These results suggest that the brain's "investment" in SST-mediated gating is a constrained optimization strategy. The system only buys the hyperbolic state when the structural hierarchy is present to guarantee a positive energy ROI.

5 Proposed Experimental Framework

We propose the **"Density vs. Topology"** Test to empirically falsify CAH, using holographic 2-photon stimulation of SST cells in Macaque PFC.

- **Control:** Normal hierarchical task performance.

- **Condition A (Scrambling):** Maintain the exact mean firing rates and synaptic count, but randomize the spatial/temporal patterns of inhibition.
- **Condition B (Silencing/Pruning):** Optogenetically silence a random 30% of the active dendritic population to simulate the "Geometric Collapse" seen in Figure 4.
- **Prediction:** Based on our "Topological Robustness" finding (Fig 1), we predict that **Condition A (Scrambling) will NOT abolish the hyperbolic phase transition**, and task performance will remain relatively intact. This would confirm that local synaptic density is the primary driver of the manifold.
- **Contrast:** Conversely, we predict that **Condition B (Pruning)** will induce the "Geometric Collapse," locking the network in a Euclidean/Spherical state and causing catastrophic failure in hierarchical inference tasks.

5.1 Non-Invasive Signature of Thermodynamic Failure

Building on the in vivo dynamics established in Section 4.3, if working memory is fundamentally a sustained hyperbolic geometry rather than a static localized engram, we can derive a non-invasive, macroscopic prediction regarding cognitive maintenance errors. We hypothesize that transient lapses in memory or attention in healthy subjects are not failures of classical encoding, but rather structural thermodynamic failures.

We predict that in high-density EEG or MEG recordings of subjects performing sustained working memory tasks, cognitive errors will be universally preceded by a distinct "Geometric Collapse." This would manifest as a transient collapse in frontoparietal gamma-band phase synchronization (reflecting a failure in W_1 transport capacity) immediately prior to the failed retrieval. This collapse signifies that the network has exhausted the metabolic budget required to sustain the apical-somatic coupling (γ) above the critical threshold (γ_c). Unable to afford the energetic cost of negative curvature, the network's topology snaps back to a flat, unintegrated Euclidean baseline, and the cognitive representation dissolves.

6 Discussion

The Curvature Adaptation Hypothesis (CAH) frames metabolic efficiency not simply as a static evolutionary constraint, but as a dynamic resource governed by geometry and regulated by specific biophysical actuators. By identifying the Martinotti subtype of Somatostatin (SST) interneurons as the primary biological driver for shifts in Ollivier-Ricci curvature, we propose a direct, quantifiable link between cellular-level conductance and network-level representational geometry.

The functional role of SST interneurons is highly diverse. For instance, Zhang et al. (2024) [25] detail a disinhibitory (SST→PV) circuit that gates hippocampal input via global gain control. Importantly, they distinguish this mechanism from the localized, targeted dendritic inhibition executed by superficial SST populations, particularly Martinotti cells.

In the CAH framework, this targeted dendritic shunting is not merely modulating gain——it is actively routing information. We formalize this as a mechanism of direct geometric control: local inhibitory dynamics physically reshape the effective topological space through which signals propagate. This provides a structural, spatial complement to existing models of flexible routing based on transient temporal synchrony [21]. Rather than relying exclusively on temporal coordination, the cortex actively warps its underlying geometry, carving out highly efficient, protected pathways for hierarchical information flow.

6.1 The Metabolic Trade-off: Local Cost for Global Efficiency

Our numerical models indicate that the brain is not statically configured for hyperbolic representation. Instead, it transiently unlocks regimes of highly negative curvature via targeted SST-mediated modulation. When apical-somatic conductance is heavily shunted ($\gamma \to 1$), the network incurs a strict local metabolic cost—the energy required to maintain steep electrochemical gradients against active inhibition.

However, this local investment purchases massive global signaling efficiency. In geometries with deep negative curvature, geodesic paths separate complex representations with exponentially fewer intermediate steps, drastically reducing the total number of action potentials required to execute complex inference. The system explicitly trades local control energy for macroscopic routing efficiency. While the exact quantitative boundaries of this trade-off in biological wetware remain an empirical frontier, the CAH provides the first concrete, mathematical bridge linking metabolism, geometry, and cognition.

6.2 Topological Robustness vs. Geometric Collapse

This geometric lens fundamentally recasts our understanding of network pathology. Conditions characterized by atypical wiring—such as cortical dysplasia or dyslexia—may scramble local structural connectivity while leaving the network's overarching geometric capacity intact. This topological robustness explains why high-level cognitive function is often preserved despite severe microstructural variation.

Conversely, conditions driven by widespread synaptic pruning, such as Alzheimer's disease, actively destroy the network's capacity to sustain the tension of negative curvature. As essential edges are lost, the system cannot support the hyperbolic plunge, effectively collapsing into a flat, Euclidean regime (see Fig. 4).

We emphasize that the pathological regimes modeled here are intentionally abstracted. Real *in vivo* degeneration involves cascading metabolic, glial, and molecular failures not captured by Optimal Transport or SNNs alone. However, the objective of the CAH is not to simulate neurochemistry, but to isolate the foundational geometric baseline through which all diverse biological disruptions ultimately manifest at the network level.

6.3 The Thermodynamic Nature of Forgetting

This geometric framework suggests a reinterpretation of cognitive loss. Classical neuroscience frequently models forgetting as the degradation or overwriting of a localized data token. However, if the integration of complex context is bound entirely within the dynamic topology of the network, then the memory is the geometry. Within this framework, forgetting corresponds to the thermodynamic inability to hold a high-energy topological shape.

These dynamics suggest a broader hypothesis: that the principles governing working memory may extend to macroscopic conscious awareness. While speculative, this framing provides a candidate mechanism that aligns with recent empirical observations of large-scale connectivity under anesthesia. For example, propofol-induced loss of consciousness is marked by the specific dissolution of alpha-band parietal-occipital connectivity [15], severing the brain's capacity for global integration.

We hypothesize that whether a chemical agent artificially severs this integration to suspend consciousness, or a natural metabolic deficit causes a frontoparietal loop to drop and extinguish a thought, both phenomena may represent a shared vulnerability to geometric collapse. Under the Curvature Adaptation Hypothesis, we propose that large-scale cognitive integration—from working memory to conscious awareness—could be governed by a singular biophysical constraint: the continuous thermodynamic capacity to sustain negative curvature within the cortical manifold.

6.4 Implications for Neuromorphic Engineering

Beyond biological wetware, these mechanics offer a radical blueprint for artificial intelligence. If curvature is a tunable thermodynamic resource, then adaptive geometry provides a physically grounded alternative to the massive, energy-dense Euclidean architectures dominating current machine learning. Mechanisms analogous to SST-mediated gating would allow artificial networks to dynamically expand and collapse their representational space in real-time, matching the precise hierarchical demands of the task.

Ultimately, this demands a paradigm shift toward neuromorphic hardware designed natively around non-Euclidean principles. If geometric adaptation effectively reduces the number of irreversible operations required, thereby mitigating the practical impact of the Landauer limit, embedding these physical phase transitions directly into silicon offers unprecedented efficiency gains. We formally define these physical implementations, including Dynamically Gated Analog Crossbars and localized thermodynamic learning rules, in companion work [23, 22].

7 Conclusion

The Curvature Adaptation Hypothesis (CAH) formulates a predictive model of a thermodynamic economy that governs biological intelligence. By expending local control energy via SST-mediated dendritic gating, the brain unlocks global hyperbolic geodesics that drastically minimize the metabolic cost of hierarchical inference. Consequently, the cortex's metabolic efficiency is not merely an anatomical byproduct, but a dynamic, density-dependent geometric capacity——one that robustly survives local variations in wiring but catastrophically fails under systemic synaptic loss.

Our numerical simulations demonstrate that cognitive health is fundamentally defined by the fluidity of this functional manifold. Specifically, we model:

- **The Healthy Regime** is characterized by a fluid, tunable phase transition, allowing the system to rest in a flat Euclidean baseline while accessing deep hyperbolic efficiency strictly on demand.
- **The Manic Regime**, simulated through high-centrality hub nodes, results in "Geometric Inelasticity". The network is forced into permanent, structural hyperbolicity, physically mandating the indiscriminate, hyper-associative states observed in psychosis.
- **The Neurodegenerative Regime**, simulated through synaptic pruning, triggers "Geometric Collapse." The system becomes trapped in a flat Euclidean manifold, physically incapable of reaching the negative curvature required for complex hierarchical integration.

Ultimately, effective information geometry is not a static anatomical feature; it is a dynamic, navigable physical state. By modeling the SST interneuron as a biophysical geometric actuator, we provide a falsifiable, mathematically rigorous framework for understanding how the brain actively warps its own representational space to metabolically survive in a highly structured, hierarchical world.

8 Data and Code Availability

The optimal transport algorithms, finite-size scaling data, and the PyNEST spiking neural network simulation script (`biological_manifold.py`) used to generate the figures in this manuscript are open-source and available in the dendritic-curvature-adaptation repository at:
https://github.com/MPender08/dendritic-curvature-adaptation

9 Acknowledgments

The author declares no competing interests.

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A Simulation Methods

Simulations were conducted in Python using the POT (Python Optimal Transport) library[5] and NetworkX[8]. Hierarchical structures were generated as stochastic Galton-Watson trees with Poisson-distributed branching (mean $b = 3$) to approximate the structural heterogeneity of biological dendrites.

Null Model Generation: To ensure that differences in curvature were due to global topology rather than local node degree, null models were generated using the **Configuration Model** approach. We employed `nx.connected_double_edge_swap` (performing $5 \times |E|$ swaps) to randomize connections while preserving the exact degree sequence of the original tree.

Computing Curvature: Transition kernels μ were computed for a sweep of $\gamma \in [0, 0.95]$. The 1-Wasserstein distance was solved using the Earth Mover's Distance algorithm over the all-pairs shortest-path distance matrix of the graph.

A.1 SNN Microcircuit Parameters

The Spiking Neural Network (SNN) was simulated using the NEST simulator[7]. The network consisted of 280 total neurons utilizing the standard exponential leaky integrate-and-fire model (`iaf_psc_exp`) with default NEST membrane parameters ($V_{th} = -55.0$ mV, $V_{reset} = -70.0$ mV). The microcircuit was partitioned into three populations: Pyramidal Cells ($N = 200$), SST interneurons ($N = 40$), and VIP interneurons ($N = 40$).

Input and Stimulation: Baseline cortical activity was simulated using Poisson generators. Pyramidal cells received background noise at 1500 Hz (weight: 500.0 pA), while the SST gate was held open by a baseline drive of 2500 Hz (weight: 600.0 pA). The macroscopic topological transition was initiated via a step-current generator targeting the VIP population, stepping from 0.0 to 1500.0 pA precisely at $t = 500$ ms.

Synaptic Connectivity:

- **Pyr $\rightarrow$ Pyr (Geodesic Highway):** Fixed in-degree connectivity (in-degree = 20), excitatory weight = 100.0 pA.
- **SST $\rightarrow$ Pyr (The Gate):** All-to-all connectivity, strongly inhibitory weight = -1000.0 pA.
- **VIP $\rightarrow$ SST (The Trigger):** All-to-all connectivity, strongly inhibitory weight = -1500.0 pA.

B Derivation of Branching Sensitivity and the OT Threshold

To understand the relationship between conductance (γ) and curvature (κ), we consider an edge (i, j) in a b-regular tree. The Wasserstein distance $W_1(\mu_i, \mu_j)$ measures the cost of transporting the mass of the lazy random walk from node i to node j.

B.1 The "Lazy" Euclidean Regime (Low Conductance)

When γ is moderate ($\gamma < \gamma_c$), the optimal transport plan exploits the local overlap between distributions. Specifically, μ_i places mass $\frac{\gamma}{b+1}$ directly onto node j, and μ_j places mass $\frac{\gamma}{b+1}$ onto node i. The Earth Mover's Distance algorithm utilizes this overlap to minimize transport cost, keeping $W_1(\mu_i, \mu_j) \approx d_G(i, j)$. Consequently, the curvature remains effectively Euclidean:

$$\kappa \approx 0 \quad \text{for} \quad \gamma < \gamma_c \tag{5}$$

B.2 The Hyperbolic Phase Transition (High Conductance)

The linear approximation represents the geometric limit where local overlap is exhausted. As $\gamma \to 1$, the "lazy" self-loop mass $(1-\gamma)$ vanishes, forcing the transport plan to shift mass between disjoint neighbor sets (distance ≥ 2).

It is only in this high-conductance regime that the curvature follows the branching-dependent scaling:

$$\lim_{\gamma \to 1} \kappa \approx \gamma \left(\frac{1-b}{b+1} \right) \tag{6}$$

This distinct switch in transport strategy——from exploiting local overlap to traversing disjoint branches——mathematically defines the phase transition observed at $\gamma_c \approx 0.78$.

B.3 Topological Robustness in Scrambled Networks

Contrary to the expectation that cycles would abolish hyperbolicity, the Scrambled Null Model retains the phase transition. This suggests that the transport cost W_1 is dominated by the local fan-out of the transition kernel μ_i. Even in a randomized graph, if the local degree k is preserved and small-world shortcuts are present, the mass dispersal at high γ remains efficient enough to approximate tree-like transport costs, driving $\kappa < 0$.